Visus: Text-Guided Face Occlusion Segmentation using CLIP-based Cross-Modal Attention Fusion

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Abstract

Segmentation of facial occlusions (e.g., glasses and face masks) is a key challenge in building robust face analysis and face detection systems. Existing approaches are mostly vision-only, which limits their ability to capture high-level semantic concepts that are naturally expressed in language. In this work, we introduce a cross-modal attention framework that integrates visual and textual representations to improve fine-grained occlusion segmentation.

Our approach leverages a pre-trained CLIP (Contrastive Language–Image Pre-training) model to bridge the gap between pixel-level visual features and semantic descriptions that come from text. Specifically, we reinterpret transformer attention maps as spatial priors and align them with text-derived representations, which helps the model localise semantically meaningful regions corresponding to occlusions. We propose a Cross-Modal Attention Fusion mechanism that facilitates structured interaction between visual attention and language embeddings, allowing the model to refine predictions using complementary cross-modal cues. This design improves the model's ability to segment the image in challenging cases where visual signals alone are insufficient.

1 Introduction

Face analysis is a fundamental component of applications such as biometric authentication and surveillance. A persistent challenge in these systems is handling facial occlusions, where regions of the face are partially covered by objects such as eyeglasses or masks. Such occlusions corrupt the feature representations extracted by face encoders and can significantly degrade recognition performance. Accurately localising occluded regions as binary masks can therefore serve as an important pre-processing step for improving robustness.

Most existing approaches treat occlusion segmentation as a purely visual task. However, this overlooks the fact that occlusions are inherently semantic: categories such as "glasses" or "face mask" carry strong priors about shape, appearance, and spatial location. These priors are naturally encoded in language and can complement visual features, especially in settings with limited supervision. Recent vision–language models such as CLIP [1] learn aligned representations across modalities, providing a natural foundation for incorporating such semantic understanding into dense prediction tasks.

In this work, we propose a framework that uses vision–language alignment to improve facial occlusion segmentation. Our key idea is to reinterpret internal attention representations from a pre-trained multimodal model as spatial priors, and to refine them through structured interaction with text-derived features. This enables the model to incorporate high-level semantic guidance while preserving fine-grained spatial detail.

2 Related Work

Transformer-based segmentation. Early work on image segmentation relied on fully convolutional networks (FCNs) [6], which enabled pixel-wise prediction directly from convolutional features. This idea was improved by encoder–decoder architectures such as U-Net [5], which combine low-level and high-level features for better localisation. More recently, Vision Transformers (ViT) [2] have shown that attention-based models can match or exceed convolutional networks on visual tasks. Methods such as SETR [7] and Segmenter [12] extend Vision Transformers to segmentation by operating on sequences of patch tokens with dedicated decoders. We also use ViT-based features, but instead of building a full decoder we use CLS-token attention maps as a simple spatial cue.

Attention maps as segmentation cues. Attention maps from transformers often highlight semantically meaningful regions. DINO [3] showed that self-supervised Vision Transformers exhibit emergent object localisation, with attention maps roughly corresponding to foreground objects despite the absence of labels. Chefer et al. [8] proposed relevance-based attribution for transformers, which can be adapted to obtain spatial importance maps, and Grad-CAM [9] uses gradients to localise important regions in CNNs. Inspired by these works, we use the CLS-to-patch attention of the final transformer layer as a coarse estimate of object regions, which is then refined by a lightweight segmentation module.

Vision–language models for localisation. CLIP [1] learns a shared embedding space for images and text, enabling zero-shot classification and retrieval. CLIPSeg [4] adds a text-conditioned decoder for language-guided segmentation; GroupViT [13] and OVSeg [19] extend this to open-vocabulary segmentation, and Grounded-SAM [14] combines grounding models with segmentation systems for text-driven queries. In contrast, our method is simpler and more focused: rather than relying on heavy decoders, we fuse text features with visual attention maps using lightweight modules.

Face occlusion handling. Occlusion in face analysis has been addressed with part-based and mask-learning models [11] and mask-aware recognition [10], as well as inpainting and attention-based methods that down-weight occluded features. Our approach instead identifies occluded regions explicitly via segmentation, which can serve as a pre-processing step for downstream recognition or reconstruction.

3 Occluded Dataset Construction

No large-scale public dataset provides pixel-level annotations for common occlusions such as glasses and face masks, so we construct a synthetic dataset based on CelebA [16]. Figure 1 illustrates the pipeline.

3.1 Base images

CelebA contains over 200,000 images of 10,177 identities (about 20 images each) covering a wide range of poses and lighting conditions, making it a suitable base for realistic occlusions.

3.2 Synthetic occlusion augmentation

- **Glasses.** Eyeglasses and sunglasses are added with OpenCV. Facial landmarks (dlib, 68 points) locate the eye region, and the glasses are resized and positioned to align with the face.

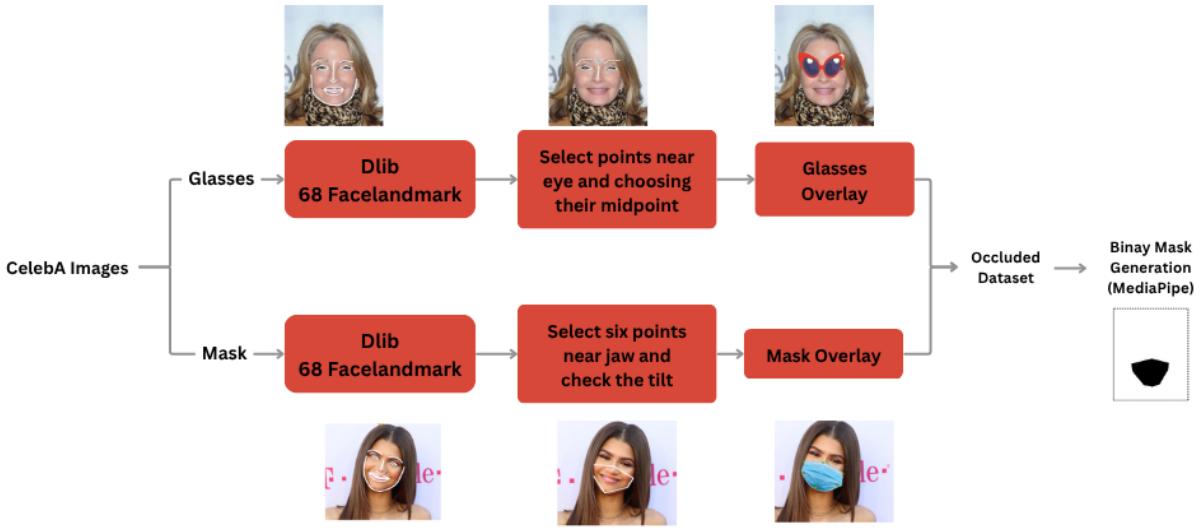


Figure 1: Synthetic occlusion dataset construction. Facial landmarks are detected on CelebA images, glasses or masks are overlaid, and the corresponding binary segmentation masks are generated.

- **Face masks.** Masks are applied with the MaskTheFace framework [17], which places realistic mask overlays based on facial landmarks so that masks conform to facial structure across poses.

3.3 Ground-truth mask generation

Ground-truth masks are produced with MediaPipe Face Mesh [18]. The detected landmarks define the occluded region (the eye region for glasses, the lower face for masks), which is rasterised into a binary mask.

3.4 Text label assignment

Each image is paired with a text label describing the occlusion. Instead of a single fixed label, each category has a small set of semantically similar prompts, and one is sampled at random for every training example to improve robustness to phrasing:

- **Glasses:** “Glasses”, “Sunglass”, “Eyewear”, “Covered eyes”, “Occluded eyes”
- **Face mask:** “Mask”, “Face mask”, “Covered face”, “Occluded region”, “Non face region”
- **Clear face:** “Face”, “Human face”, “Visible face”, “Uncovered face”, “Clear face region”

4 Method

Given an input image and a text prompt (e.g. “glasses” or “face mask”), the image is processed by the CLIP visual encoder and the prompt by the CLIP text encoder. From these two branches we extract attention-based features: the visual attention map gives a coarse localisation of relevant regions, and the text features give semantic context. The two are fused by cross-modal attention modules inside a segmentation head that produces the final mask. Figure 2 gives an overview.

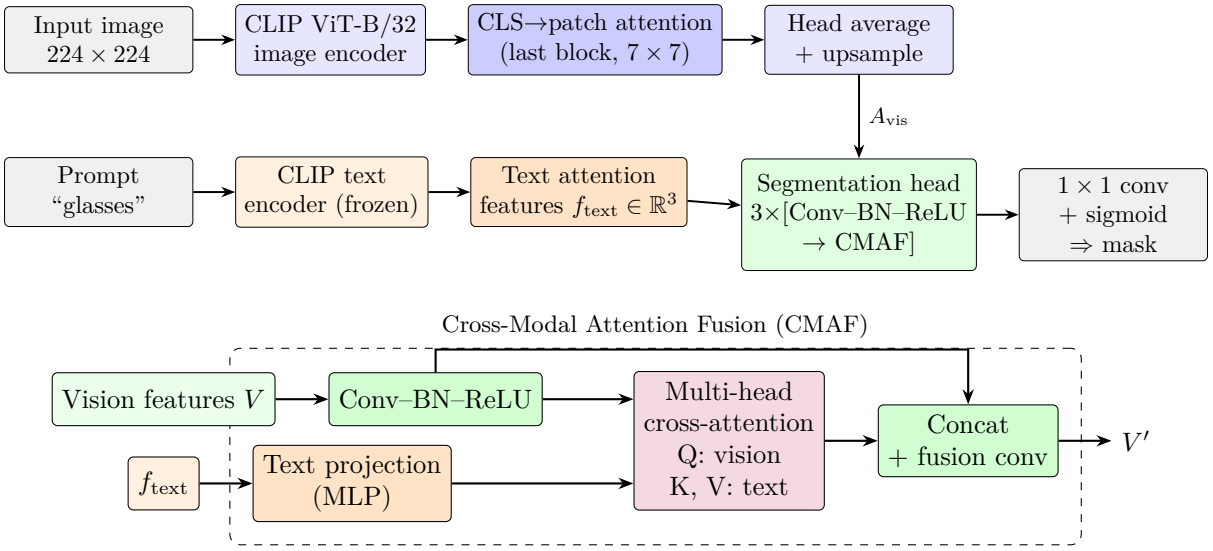


Figure 2: Top: overview of the framework. Visual attention from CLIP is combined with text-derived features through cross-modal attention to produce a segmentation mask. Bottom: one CMAF block.

4.1 Backbone design

CLIP consists of a vision encoder and a text encoder, which we treat differently.

- **Text encoder (frozen).** The text encoder is kept completely frozen. Pre-trained on large-scale image–text data, it already captures rich semantic relationships between language and visual concepts (e.g. “glasses”, “face mask”). Fine-tuning it on a small task-specific dataset risks catastrophic forgetting; freezing it preserves its ability to generalise across phrasings.
- **Vision encoder (partially fine-tuned).** Only the final transformer blocks of the vision encoder are set as trainable. Early layers capture low-level, transferable features such as edges and textures, whereas deeper layers encode higher-level semantics better suited to adaptation. This lets the model specialise while retaining the robustness of pre-trained features and reducing overfitting and compute.

4.2 Role of attention maps

CLIP embeddings are a global representation of the whole image, well suited to classification but insufficient for segmentation, which needs spatial localisation. Attention maps preserve spatial information: the CLS token attends to image patches, and the attention weights indicate the relative importance of each region. We use these maps as an initial estimate of the occluded region.

4.3 Visual attention extraction

We extract attention from the final transformer layer of the visual encoder, focusing on the interaction between the CLS token and the image patches. This gives a tensor $a \in \mathbb{R}^{B\times H\times N}$, where H is the number of heads and N the number of patches. We average over heads, reshape to a spatial grid and upsample to the input resolution:

$$A_{\text{vis}} = \text{Upsample}\left(\frac{1}{H}\sum_{h=1}^{H} a^{(h)}\right). \tag{1}$$

The resulting map is a coarse spatial prior over regions of interest.

4.4 Text feature representation

From the text encoder we extract a compact vector that summarises attention patterns across tokens. Instead of using the full embedding, we aggregate CLS-to-token attention from each of the last three layers into one scalar per layer, giving $f_{\text{text}} \in \mathbb{R}^3$. This vector acts as the semantic signal that guides segmentation.

4.5 Cross-Modal Attention Fusion

To combine visual and textual information we use a Cross-Modal Attention Fusion (CMAF) module. Conceptually, visual features indicate *where* relevant regions are, and text features indicate *what* the model should focus on. Cross-attention, with spatial vision tokens as queries and the projected text feature as key and value, lets the model refine visual attention using semantic cues:

$$V' = \text{Fusion}\left(V,\ \text{CrossAttention}(V, f_{\text{text}})\right). \tag{2}$$

4.6 Segmentation head

The head refines the visual attention map through three convolutional stages, each followed by a CMAF block (channel flow $1 \to 128 \to 64 \to 128 \to 64 \to 64 \to 64 \to 1$). Spatial resolution is kept fixed while the features are progressively refined. A final 1×1 convolution and sigmoid output a single-channel mask in $[0, 1]$.

4.7 Dice loss

In occlusion segmentation the foreground is typically much smaller than the background, and losses such as binary cross-entropy that weigh every pixel equally can perform poorly under this imbalance. Dice loss measures overlap directly:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\,TP + \epsilon}{2\,TP + FP + FN + \epsilon}. \tag{3}$$

It decreases as overlap increases, penalises both false positives and false negatives, focuses on region-level rather than per-pixel agreement, and handles class imbalance naturally.

5 Experiments

5.1 Implementation details

All experiments use CLIP ViT-B/32 from OpenAI's `clip` library. Images are pre-processed with CLIP's default pipeline (resize to 224×224, normalise); masks are resized to 224×224 and binarised at 0.5. Training runs on a single NVIDIA GPU, with checkpoints saved every two epochs. Table 1 lists the hyperparameters.

5.2 Training curves

Figure 3 shows the Dice loss of saved checkpoints across training, and Table 2 reports it at selected epochs.

5.3 Segmentation results: CelebA and LFW

We evaluate on a held-out subset of CelebA and on LFW [15] to assess generalisation to an unseen distribution, using mean Intersection over Union:

$$\text{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{|P_i \cap G_i|}{|P_i \cup G_i|}, \tag{4}$$

Table 1: Training hyperparameters.

Hyperparameter	Value
CLIP backbone	ViT-B/32
Image size	224×224
Batch size	24
Optimiser	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
Epochs	50
Train / val split	80% / 20%
Dice loss smoothing ϵ	1.0
Text layers used	3
CMAF hidden dim	64
Segmentation head hidden channels	128

Table 2: Dice loss (↓) at selected training epochs.

Epoch	Dice loss
6	0.0559
10	0.0468
18	0.1650
22	0.0633
24	0.0383
50	0.0336

where P_i and G_i are the predicted and ground-truth binary masks for sample i. Ground-truth masks for evaluation are generated with MediaPipe facial landmarks. Table 3 reports the scores.

Table 3: mIoU on held-out test sets.

Dataset	mIoU	Notes
CelebA (held-out)	0.9441	In-distribution
LFW	0.9405	Out-of-distribution

The model achieves an mIoU of 0.9441 on held-out CelebA and 0.9405 on LFW. The small gap between in-distribution and out-of-distribution performance suggests good generalisation, which we attribute to the frozen CLIP text encoder's semantic representations and the prompt augmentation used during training.

5.4 Qualitative analysis

Figure 4 shows example LFW inputs with the CLS-token attention map, the predicted mask and the ground truth. The raw CLIP attention is spatially diffuse and centred broadly on prominent facial features; the segmentation head reshapes it towards the occluded region named in the prompt.

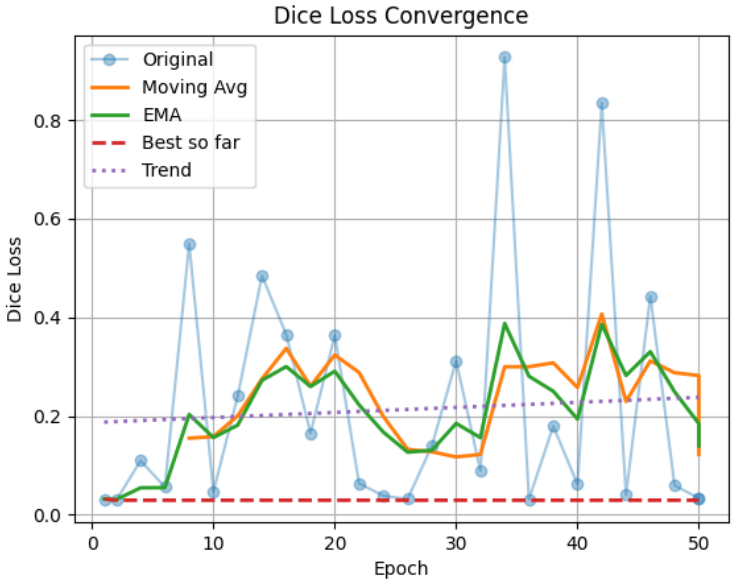


Figure 3: Dice loss across the 50 training epochs, with moving average, exponential moving average, best-so-far and linear trend.

6 Discussion

Freezing strategy. Keeping the text encoder frozen is important: it has already learned precise semantic representations of occlusion categories, and fine-tuning it on a small task-specific dataset risks degrading them.

Compact text representation. The text feature $f_{\text{text}} \in \mathbb{R}^3$ is deliberately minimal: it encodes only per-layer CLS attention statistics rather than the full 512-dimensional CLIP text embedding. This was motivated by parameter efficiency, but it is a potential bottleneck; richer text representations may improve performance.

Dice loss for class imbalance. Glasses and face masks typically cover a small fraction of the face image. Binary cross-entropy can be dominated by the majority class in this setting, whereas Dice loss directly optimises overlap between prediction and ground truth.

7 Conclusion

We presented a lightweight, language-guided face occlusion segmentation framework that repurposes the internal attention of a pre-trained CLIP model. By combining CLS-token visual attention maps with compact multi-layer text attention statistics through learned cross-modal fusion modules, the system segments glasses and face-mask occlusions without a heavy decoder or full backbone fine-tuning, and reaches an mIoU of 0.9441 on CelebA and 0.9405 on LFW.

Limitations and future work. The text representation is very compact ($\mathbb{R}^3$); richer encodings such as the full 512-dimensional CLIP text embedding may improve segmentation quality. The approach uses prompt augmentation at training time but does not explore zero-shot generalisation to unseen occlusion categories. Data augmentation (random crops, photometric jitter) and more principled prompt engineering could improve generalisation, and extending the framework to

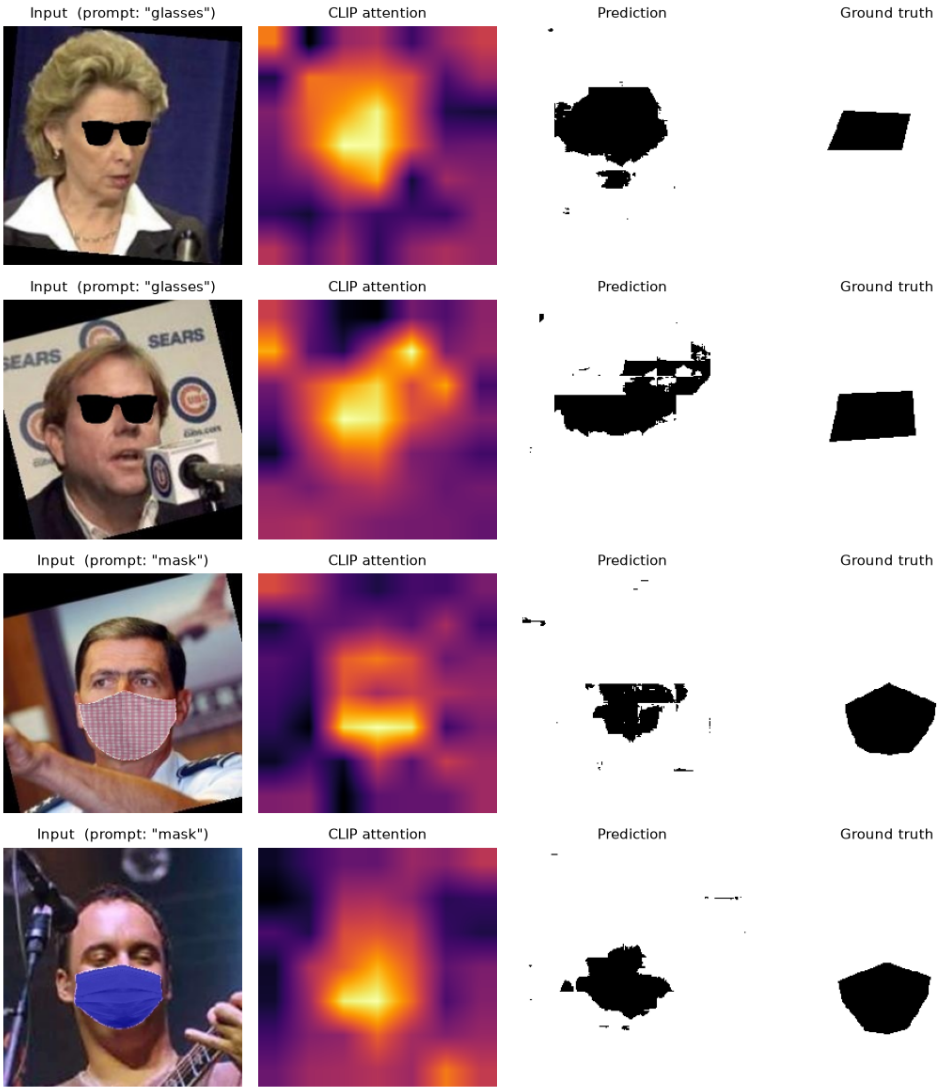


Figure 4: Qualitative results on LFW. Columns: input image with prompt, CLS-token attention map, predicted mask, ground truth (occluded region shown in black).

multi-class occlusion segmentation and evaluating on benchmarks such as the AR face database would provide broader validation.

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